RACE
Regular Analysis Through Continuous Exercise

Geometrical Optics : RACE-01

1. Red light differs from blue light as they have :
(A) Different frequencies and different wavelengths.
(B) Different frequencies and same wavelengths.
(C) Same frequencies and same wavelengths.
(D) Same frequencies and different wavelengths.

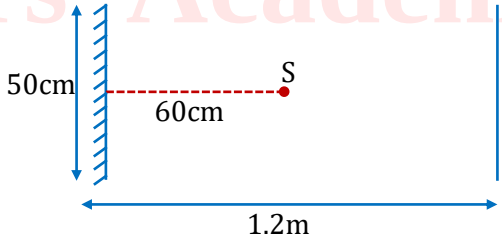
PGO276

2. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be :
(A) 90° (B) 45° (C) 75° (D) 60°

PGO277

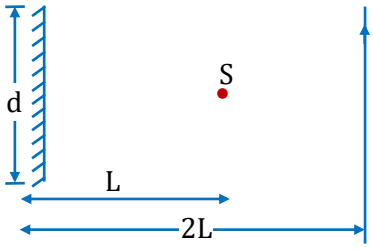
3. A point source of light *S*, placed at a distance 60 *cm* infront of the centre of a plane mirror of width 50 *cm*, hangs vertically on a wall. A man walks infront of the mirror along a line parallel to the mirror at a distance 1.2 *m* from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____ *cm*.

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PGO278

4. A point source of light, *S* is placed at a distance *L* in front of the centre of plane mirror of width *d* which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2*L* as shown below. The distance over which the man can see the image of the light source in the mirror is :

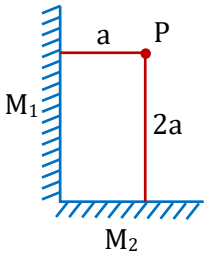


(A) 3*d* (B) $\frac{d}{2}$ (C) *d* (D) 2*d*

PGO279

For More Join: @IITJEE_Advanced

5. Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source 'P' is placed at 'a' and '$2a$' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5}$ = 2.3)



(A) 3 a (B) 4.6 a (C) 2 a (D) $2\sqrt{10}a$

PGO280

6. A man, whose height is 1.75m is standing at a distance 6m from a vertical pillar of height h = 7 m. At what distance from the man should we put a small flat mirror on the ground horizontally to see in it the image of the top of the pillar ?

(A) 4.8 m (B) 1.2 m (C) 5.2 m (D) 4.2 m

PGO281

7. The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively. Then choose the correct relation for these vectors.

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(A) $\vec{b}=\vec{a}+2\vec{c}$ (B) $\vec{b}=2\vec{a}+\vec{c}$ (C) $\vec{b}=\vec{a}-2(\vec{a}.\vec{c})\vec{c}$ (D) $\vec{b}=\vec{a}-\vec{c}$

PGO282

8. Which of the following statements about a plane mirror is (are) **CORRECT** ?

(A) When a plane mirror is displaced a distance L perpendicular to its surface, image point for any fixed object point moves a distance 2L.

(B) When a plane mirror is displaced a distance L perpendicular to its surface, image point for any fixed object point moves a distance L.

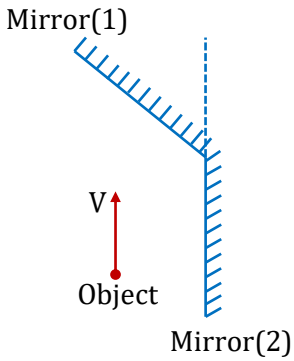
(C) When a plane mirror is rotated through an angle α about an axis that lies in the plane of the mirror, the reflected ray for any particular incident ray that is perpendicular to the axis rotates through an angle 2α.

(D) When a plane mirror is rotated through an angle α about an axis that lies in the plane of the mirror, the reflected ray for any particular incident ray that is perpendicular to the axis rotates through an angle α.

PGO283

For More Join: @IITJEE_Advanced

9. In the diagram shown, all the velocities are given with respect to earth. What is the relative velocity of the image in mirror (1) with respect to the image in the mirror (2)? The mirror (1) forms an angle β with the vertical.



(A) $2V\sin 2\beta$ (B) $2V\sin\beta$ (C) $2V/\sin 2\beta$ (D) None

PGO284

10. A particle is moving with velocity $(\hat{i}+2\hat{j}+3\hat{k})$ m/s. A plane mirror exists in the x-y plane. The mirror has a velocity $(\hat{i}+2\hat{j})$ m/s. Velocity of the image (in m/s) is:

(A) $\hat{j}+\hat{k}$ (B) $\hat{i}+2\hat{j}-3\hat{k}$ (C) $-\hat{j}-\hat{k}$ (D) None of these

PGO285

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ANSWER KEY

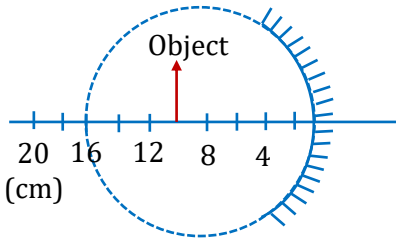
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	150	A	C	B	C	AC	B	B

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Geometrical Optics : RACE-02

1. A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)



(A) Inverted, real and magnified (B) Erect, virtual and magnified
(C) Erect, virtual and unmagnified (D) Inverted, real and unmagnified

PGO286

2. A concave mirror for face viewing has focal length of 0.4 *m*. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is :
(A) 1.60 *m* (B) 0.24 *m* (C) 0.16 *m* (D) 0.32 *m*

PGO287

3. The focal length *f* is related to the radius of curvature *r* of the spherical convex mirror by:
(A) $f = +\frac{1}{2}r$ (B) $f = -r$ (C) $f = -\frac{1}{2}r$ (D) $f = r$

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PGO288

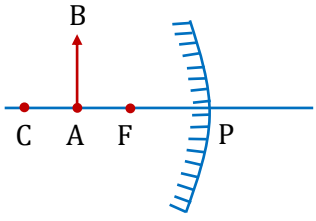
4. A short straight object of height 100 *cm* lies before the central axis of a spherical mirror whose focal length has absolute value $|f| = 40cm$. The image of object produced by the mirror is of height 25 *cm* and has the same orientation of the object. One may conclude from the information :
(A) Image is real, same side of concave mirror.
(B) Image is virtual, opposite side of concave mirror.
(C) Image is real, same side of convex mirror.
(D) Image is virtual, opposite side of convex mirror.

PGO289

5. A convex mirror of radius of curvature 30 *cm* forms an image that is half the size of the object. The object distance is :
(A) –15 *cm* (B) 45 *cm* (C) –45*cm* (D) 15 *cm*

PGO290

6. An object is kept between centre of curvature *C* and focus *F*. Image will be formed:



(A) between *F* and *C* (B) between *C* and ∞ (C) between *F* and *P* (D) None of these

PGO291

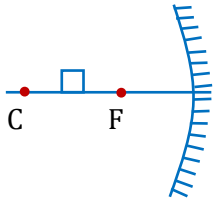
For More Join: @IITJEE_Advanced

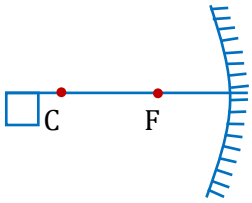
7. In a convex mirror image formed will be

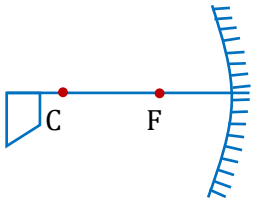
(A) Always virtual (B) Always real (C) Never virtual (D) None of these

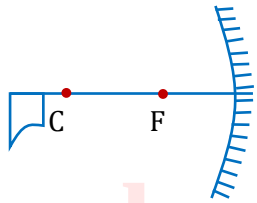
PGO292

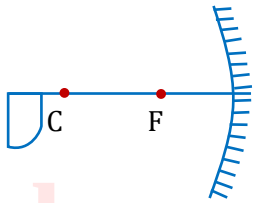
8. A small square object is placed between centre of curvature and focus of concave mirror as shown in figure. What will be the correct shape of image ?



(A) 

(B) 

(C) 

(D) 

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PGO293

9. When an object is kept at a distance of 30 *cm* from a concave mirror, the image is formed at a distance of 10 *cm* from the mirror. If the object is moved with a speed of 9 cms^{-1}, the speed (in cms^{-1}) with which image moves at that instant is _____.

PGO294

10. Car *B* overtakes another car *A* at a relative speed of 40 ms^{-1}. How fast will the image of car *B* appear to move in the mirror of focal length 10 *cm* fitted in car *A*, when the car *B* is 1.9 *m* away from the car *A*?

(A) 4 ms^{-1} (B) 0.2 ms^{-1} (C) 40 ms^{-1} (D) 0.1 ms^{-1}

PGO295

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	A	D	A	D	D	B	1	D

For More Join: @IITJEE_Advanced

Regular Analysis Through Continuous Exercise

Geometrical Optics : RACE-03

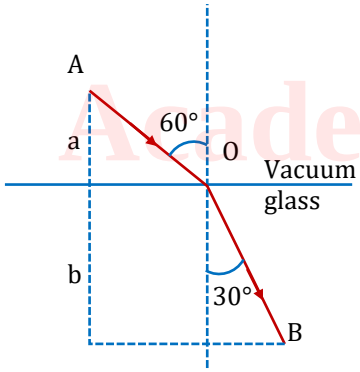
1. If a wave gets refracted into a denser medium, then which of the following is true?
(A) Wavelength, speed and frequency decreases.
(B) Wavelength increases, speed decreases and frequency remains constant.
(C) Wavelength and speed decreases but frequency remains constant.
(D) Wavelength, speed and frequency increases.

PGO296

2. The difference of speed of light in the two media A and B ($v_A - v_B$) is $2.6 \times 10^7\ m/s$. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is :
(Given : speed of light in vacuum $c = 3 \times 10^8\ ms^{-1}$)
(A) 1.303 (B) 1.318 (C) 1.13 (D) 0.12

PGO297

3. A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is:



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(A) $2a + 2b$ (B) $2a + \frac{2b}{3}$ (C) $\frac{2\sqrt{3}}{a} + 2b$ (D) $2a + \frac{2b}{\sqrt{3}}$

PGO298

4. A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water ($\mu = 4/3$) into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is actually filled ?
(A) 11.7 cm (B) 10 cm (C) 7.5 cm (D) 8.75 cm

PGO299

5. A vessel of depth $2h$ is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be :
(A) $\frac{h}{\sqrt{2}}$ (B) $\frac{3}{4}h\sqrt{2}$ (C) $\frac{h}{2\left(\sqrt{2}+1\right)}$ (D) $\frac{h}{3\sqrt{2}}$

PGO300

For More Join: @IITJEE_Advanced

6. The critical angle of a medium for a specific wavelength, if the medium has relative permittivity 3 and relative permeability $\frac{4}{3}$ for this wavelength, will be :

(A) 60° (B) 15° (C) 45° (D) 30°

PGO301

7. A ray of laser of a wavelength 630 *nm* is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option:

(A) Angle of refraction is 24.41° (B) Angle of refraction is 30°

(C) Refraction is not possible (D) Angle of refraction is 53.4°

PGO302

8. A light wave travelling linearly in a medium of dielectric constant 4, incident on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be

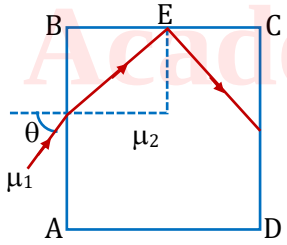
(Given : relative permeability of medium $\mu_r = 1$)

(A) 10° (B) 20° (C) 30° (D) 60°

PGO303

9. A transparent cube of side *d*, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1(\mu_1 < \mu_2)$. A ray is incident on the face *AB* at an angle θ(shown in the figure). Total internal reflection takes place at point *E* on the face *BC*. The θ must satisfy :

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(A) $\theta < \sin^{-1}\frac{\mu_1}{\mu_2}$ (B) $\theta < \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$ (C) $\theta > \sin^{-1}\frac{\mu_1}{\mu_2}$ (D) $\theta > \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$

PGO304

10. Which of the following statement is correct ?

(A) In primary rainbow, observer sees red colour on the top and violet on the bottom.

(B) In primary rainbow, observer sees violet colour on the top and red on the bottom.

(C) In primary rainbow, light wave suffers to internal reflection twice before coming out of water drops.

(D) Primary rainbow is less bright than secondary rainbow.

PGO305

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	A	B	B	D	C	D	B	A

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Geometrical Optics : RACE-04

1. A light ray enters a solid glass sphere of refractive index $\mu=\sqrt{3}$ at an angle of incidence 60°. The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degrees) between the reflected and refracted rays at this surface is_____.

PGO306

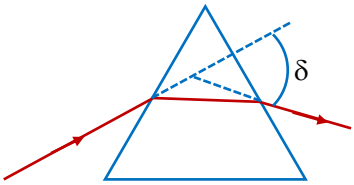
2. The angle of deviation through a prism is minimum when:

(a) Incident ray and emergent ray are symmetric to the prism.

(b) The refracted ray inside the prism becomes parallel to its base.

(c) Angle of incidence is equal to that of the angle of emergence .

(d) When angle of emergence is double the angle of incidence.



Choose the correct answer from the options given below :

(A) Statements (a), (b) and (c) are true. (B) Only statement (d) is true.

(C) Only statements (a) and (b) are true. (D) Statements (b) and (d) are true.

PGO307

3. A ray of light passing through a prism ($\mu = \sqrt{3}$) suffers minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. Then, the angle of prism is _____ (in degrees)

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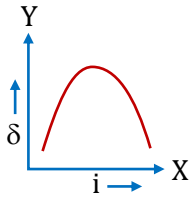
PGO308

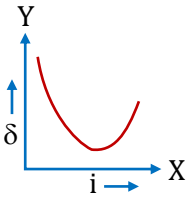
4. A prism of refractive index μ and angle of prism *A* is placed in the position of minimum angle of deviation. If minimum angle of deviation is also *A*, then in terms of refractive index, *A* is

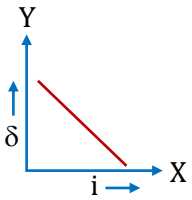
(A) $2\cos^{-1}\left(\frac{\mu}{2}\right)$ (B) $\sin^{-1}\left(\frac{\mu}{2}\right)$ (C) $\sin^{-1}\left(\sqrt{\frac{\mu-1}{2}}\right)$ (D) $\cos^{-1}\left(\frac{\mu}{2}\right)$

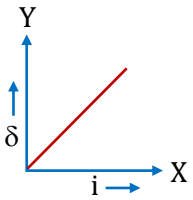
PGO309

5. The expected graphical representation of the variation of angle of deviation 'δ' with angle of incidence '*i*' in a prism is

(A)


(B)


(C)


(D)


PGO310

For More Join: @IITJEE_Advanced

6. A prism of angle $A = 1°$ has a refractive index $\mu = 1.5$. A good estimate for the minimum angle of deviation (in degrees) is close to $N/10$. Value of N is _____.

PGO311

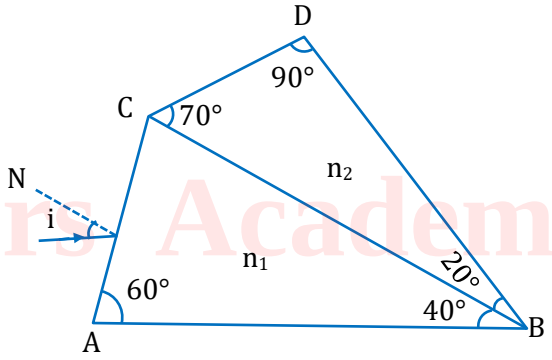
7. The image of an object placed in air formed by a convex refracting surface is at a distance of 10 *m* behind the surface. The image is real and is at $2/3^{rd}$ of the distance of the object from the surface. The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$ *m*. The value of 'x' is_____.

PGO312

8. A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation:

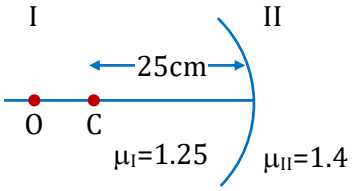
$$n_1 = 1.2 + \frac{10.8\times10^{-14}}{\lambda^2} \text{ and } n_2 = 1.45 + \frac{1.8\times10^{-14}}{\lambda^2}$$

The wavelength for which rays incident at any angle on the interface *BC* pass through without bending at that interface will be _____ *nm*.



PGO313

9. Region I and II are separated by a spherical surface of radius 25 *cm*. An object is kept in region I at a distance of 40 *cm* from the surface. The distance of the image from the surface is :



(A) 55.44 *cm* (B) 9.52 *cm* (C) 18.23 *cm* (D) 37.58 *cm*

PGO314

10. Light from a point source in air falls on a convex curved surface of radius 20 *cm* and refractive index 1.5. If the source is located at 100 *cm* from the convex surface, the image will be formed at _____ *cm* from the object.

PGO315

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	90	A	60	A	B	5	30	600	D	200

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Geometrical Optics : RACE-05

1. The refractive index of a converging lens is 1.4. What will be the focal length of this lens if it is placed in a medium of same refractive index ?
(Assume the radii of curvature of the faces of lens are R_1 and R_2 respectively)

(A) 1 (B) Infinite (C) $\frac{R_1R_2}{R_1-R_2}$ (D) Zero

PGO316

2. An object is placed at the focus of concave lens having focal length f. What is the magnification and distance of the image from the optical centre of the lens?

(A) $1, \infty$ (B) Very high, ∞ (C) $\frac{1}{2}, \frac{f}{2}$ (D) $\frac{1}{4}, \frac{f}{4}$

PGO317

3. A convex lens of focal length 20 *cm* produces images of the same magnification 2 when an object is kept at two distances x_1 and x_2 ($x_1 > x_2$) from the lens. The ratio of x_1 and x_2 is

(A) 5 : 3 (B) 2 : 1 (C) 4 : 3 (D) 3 : 1

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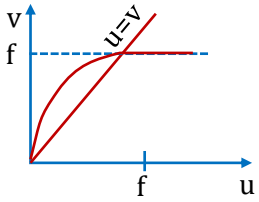
PGO318

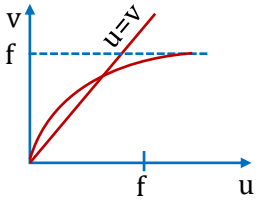
4. A double convex lens has power P and same radii of curvature R of both the surfaces. The radius of curvature of a surface of a plano-convex lens made of the same material with power 1.5 P is:

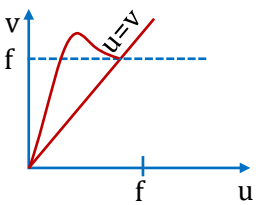
(A) $\frac{R}{2}$ (B) $2R$ (C) $\frac{3R}{2}$ (D) $\frac{R}{3}$

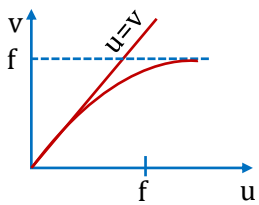
PGO319

5. For a concave lens of focal length f, the relation between object and image distance u and v, respectively, from its pole can best be represented by ($u = v$ is the reference line):

(A)


(B)


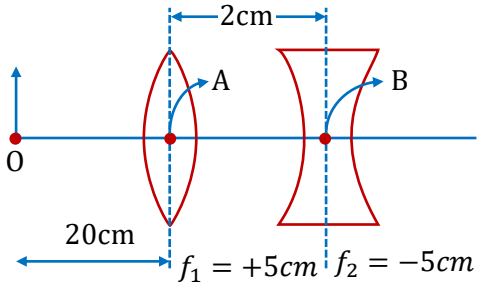
(C)


(D)


PGO320

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6. What is the position and nature of image formed by lens combination shown in figure?
(f_1, f_2 are focal lengths)



(A) 70 *cm* from point *B* at left; virtual
(B) 40 *cm* from point *B* at right; real
(C) $\frac{20}{3}$ *cm* from point *B* at right, real
(D) 70 *cm* from point *B* at right, real

PGO321

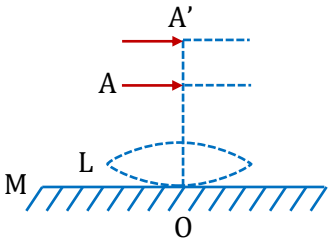
7. A point like object is placed at a distance of 1*m* in front of a convex lens of focal length 0.5 *m*. A plane mirror is placed at a distance of 2 *m* behind the lens. The position and nature of the final image formed by the system is :

(A) 1 *m* from the mirror, virtual
(B) 1 *m* from the mirror, real
(C) 2.6 *m* from the mirror, real
(D) 2.6 *m* from the mirror, virtual

PGO322

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8. A thin convex lens *L* (refractive index = 1.5) is placed on a plane mirror *M*. When a pin is placed at *A*, such that *OA* = 18 *cm*, its real inverted image is formed at *A* itself, as shown in figure. When a liquid of refractive index μ_1 is put between the lens and the mirror. The pin has to be moved to *A*', such that *OA*' = 27 *cm*, to get its inverted real image at *A*' itself. The value of μ_1 will be :-



(A) $\sqrt{2}$
(B) $\frac{4}{3}$
(C) $\sqrt{3}$
(D) $\frac{3}{2}$

PGO323

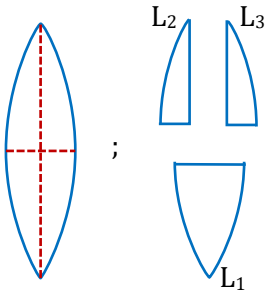
9. An object is at a distance of 20 *m* from a convex lens of focal length 0.3 *m*. The lens forms an image of the object. If the object moves away from the lens at a speed of 5 *m/s*, the speed and direction of the image will be:

(A) 0.92×10^{-3} *m/s* away from the lens
(B) 2.26×10^{-3} *m/s* away from the lens
(C) 1.16×10^{-3} *m/s* towards the lens
(D) 3.22×10^{-3} *m/s* towards the lens

PGO324

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10. A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figure). Choose the incorrect option for the reported pieces.



(A) Power of $L_1 = \frac{P}{2}$ (B) Power of $L_2 = \frac{P}{2}$ (C) Power of $L_3 = \frac{P}{2}$ (D) Power of $L_1 = P$

PGO325

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	D	D	D	D	C	B	C	A

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RACE
Regular Analysis Through Continuous Exercise

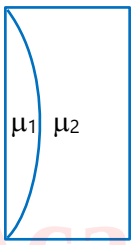
Geometrical Optics : RACE-06

1. A plano-convex lens (focal length f_2, refractive index μ_2, radius of curvature R) fits exactly into a plano-concave lens (focal length f_1, refractive index μ_1, radius of curvature R). Their plane surfaces are parallel to each other. Then, the focal length of the combination will be :

(A) $f_1 - f_2$ (B) $f_1 + f_2$ (C) $\frac{R}{\mu_2 - \mu_1}$ (D) $\frac{2f_1f_2}{f_1 + f_2}$

PGO326

2. Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.

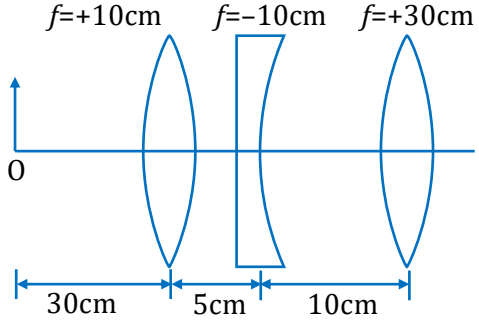


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(A) $\frac{1}{\mu_2 - \mu_1}$ (B) $\mu_1 - \mu_2$ (C) $\frac{1}{\mu_1 - \mu_2}$ (D) $\mu_2 - \mu_1$

PGO327

3. Find the distance of the image from object *O*, formed by the combination of lenses in the figure :



(A) 75 *cm* (B) 10 *cm* (C) 20 *cm* (D) Infinity

PGO328

4. Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as :

(A) Astigmatism (B) Myopia with Astigmatism
(C) Presbyopia with Astigmatism (D) Myopia and hypermetropia

PGO329

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5. The thickness at the centre of a plano convex lens is 3 *mm* and the diameter is 6 *cm*. If the speed of light in the material of the lens is 2×10^8 ms^{-1}. The focal length of the lens is _____.

(A) 0.30 *cm* (B) 15 *cm* (C) 1.5 *cm* (D) 30 *cm*

PGO330

6. A compound microscope consists of an objective lens of focal length 1*cm* and an eye piece of focal length 5 *cm* with a separation of 10 *cm*. The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ *cm*. The value of n is ___.

PGO331

7. In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 *cm*. The focal length of the objective, when the angular magnification of the telescope is 2, will be:

(A) 20 *cm* (B) 30 *cm* (C) 10 *cm* (D) 15 *cm*

PGO332

8. The magnifying power of a telescope with tube 60 *cm* is 5. What is the focal length of its eye piece?

(A) 30 *cm* (B) 40 *cm* (C) 20 *cm* (D) 10 *cm*

PGO333

9. A microscope was initially placed in air (refractive index 1). It is then immersed in oil (refractive index 2). For a light whose wavelength in air is λ, calculate the change of microscope's resolving power due to oil and choose the correct option:

(A) Resolving power will be $\frac{1}{4}$ in the oil than it was in the air.

(B) Resolving power will be twice in the oil than it was in the air.

(C) Resolving power will be four times in the oil than it was in the air.

(D) Resolving power will be $\frac{1}{2}$ in the oil than it was in the air.

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PGO334

10. The aperture of the objective is 24.4 *cm*. The resolving power of this telescope. If a light of wavelength 2440 Å is used to see the object will be

(A) 8.1×10^6 (B) 10.0×10^7 (C) 8.2×10^5 (D) 1.0×10^{-8}

PGO335

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	A	B	D	50	A	D	B	C

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